Skills Summary

Getting the Centre, the Vector and the Factor Right

Use this reference while completing your worksheet or when studying.

Three things a transformation problem can get wrong, and they fail differently. A wrong *vector* sends the figure the wrong way. A wrong *centre* gives the right size in the wrong place. A wrong *scale factor* gives the wrong size. Naming which one broke is half the diagnosis.

Skill 1 — The translation vector, in the right order

Rule: the vector from $P(x, y)$ to $P'(p, q)$ has components $p - x$ and $q - y$ — *image minus original*, in that order.

Reversing the subtraction negates both components, which points the translation exactly backwards. Check by applying your rule to the original point and seeing whether it lands on the image.

Example: A translation maps $(6, 2)$ to $(-1, 5)$. Find its components.
Step 1. The vector runs from the original point to the image, so each component is the image coordinate minus the original one, not the other way round.
Step 2. horizontal: $-1 - 6$ and vertical: $5 - 2$
$\Rightarrow$ ($\mathbf{-7}$, $\mathbf{3}$)

Try it: A translation maps $(9, 4)$ to $(2, 4)$. Give its horizontal component.

check: -7

Skill 2 — Dilating about the given centre

Rule: about a centre (h, j) with factor k, $(x, y) \to \big(h + k(x - h),\ \ j + k(y - j)\big)$.

Only when the centre is the origin does this collapse to $(kx,\ ky)$. Using (kx, ky) with any other centre produces a figure of the correct size in the wrong position — and the centre is the one point that must not move.

Example: Dilate $(8, 4)$ about $(3, 1)$ by a factor of 2.
Step 1. The centre is not the origin, so scale each coordinate's offset from the centre and add it back — multiplying the raw coordinates would dilate about the origin instead.
Step 2. $x' = 3 + 2(8 - 3) = 3 + 10$ and $y' = 1 + 2(4 - 1) = 1 + 6$
$\Rightarrow$ ($\mathbf{13}$, $\mathbf{7}$)

Try it: Dilate $(7, 2)$ about $(5, 2)$ by a factor of 4 and give the x-coordinate.

check: **13**

Skill 3 — Getting the scale factor from the right ratio

From lengths: $k = \dfrac{\text{image length}}{\text{original length}}$ — a quotient, never a difference, and a pure number with no unit.

From areas: the area ratio equals k^2, so $k = \sqrt{\text{area ratio}}$. Handing the area ratio straight over as k is the commonest slip in this topic.

Example: Two similar figures have areas 36 and 100. Find the linear scale factor.
Step 1. The ratio given is a ratio of areas, and areas scale by the square of the linear factor, so the ratio is already k squared — take a square root before using it.
Step 2. $k^2 = 100/36$, so $k = \sqrt{100/36} = 10/6$
$\Rightarrow \quad k \approx \mathbf{1.67}$

Try it: Two similar figures have areas 20 and 45. Find the linear scale factor.

check: **1.5**